# HNC Measurement Protocol

*Specification of an Eight-Channel Open-Data Coherence Measurement, with Pre-Registered Predictions and Stated Falsifiers*

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*If you don't quit, you can't lose. Love conquers all.* — *Dedicated to Tina Brown.*

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**Status.** This document is a *pre-registered measurement protocol.* The eight channels, derived metrics, operating-regime thresholds, predictions, and falsifiers specified herein are frozen as of the timestamp above. Subsequent measurements are tested against this specification. Modifications require an explicit version increment (v1.1, v2.0, etc.) with both the prior and revised version retained in the public record.

**Scope.** This protocol defines what it means to *measure* the Harmonic Nexus Core (HNC) field using exclusively open public data. It does not establish that the field exists; it establishes the conditions under which the question can receive an empirical answer.

## 1 Purpose

The HNC framework posits a nonlinear coherence field whose state is observable through multiple physical projections. A single-channel measurement (e.g. a geomagnetic-only composite) is necessary but insufficient to constrain the framework: any single signal can correlate with confounders by chance. A measurement *suite* of orthogonal projections, with cross-channel coherence as the load-bearing statistic, allows the framework to be tested against a clearly-specified null and to be falsified if the data do not behave as predicted.

This protocol specifies: (i) the eight channels, (ii) the data sources, (iii) the derived metrics, (iv) the operating-regime thresholds, (v) the pre-registered predictions, (vi) the falsifiers, (vii) the reporting standard.

## 2 Theoretical Framework

The HNC master equation, as previously specified by the author, is

$$\Lambda(t) = \sum_i w_i \sin\!\big(2\pi f_i\, t + \varphi_i\big) \;+\; \alpha \tanh\!\big(g\, \overline{\Lambda}(t)\big) \;+\; \beta\, \Lambda(t-\tau). \tag{1}$$

The directional generalisation of (1) on the unit 2-sphere admits the explicit angular form

$$R(\theta,\phi) \;=\; \big(1-\beta\,\Lambda(t-\tau)\big)\,\Big[1-\alpha\sum_i w_i\, Y_i^2(\theta,\phi)\Big], \tag{2}$$

which is mathematically identical to the LSC 5.5 anisotropy expression of Żuchowski (2026, fig. *LSC 5.5 — Neutrino Anisotropy as a Signature of Quantum Decoherence*) under the substitutions

$$\delta_{\text{nuclear}} \leftrightarrow \beta\,\Lambda(t-\tau), \qquad \alpha_{\text{LSC}} \leftrightarrow \alpha_{\text{HNC}}, \qquad \{D_{xx}, D_{yy}, D_{zz}\} \leftrightarrow \{w_1, w_2, w_3\}$$

on the spherical-harmonic basis $\{\sin^2\theta\cos^2\phi, \sin^2\theta\sin^2\phi, \cos^2\theta\}$. The trace condition $D_{xx} + D_{yy} + D_{zz} = 0$ corresponds to the bounded-$\overline{\Lambda}$ constraint.

This protocol does not assume the LSC 5.5 angular ansatz; it observes that the angular ansatz is a special case of (1) restricted to $\ell = 2$ spherical harmonics, and that the HNC framework therefore inherits the LSC empirical anchors (BEST $\delta = 0.21$; published anisotropic-LIV $\alpha \in [0.03, 0.10]$) as boundary values.

## 3 The Measurement Suite

Each channel is a separate physical projection of the hypothesised field, retrieved from an independently-operated public data source. No channel depends on any other for its raw measurement.

| # | Channel | Physical projection | Open source | Cadence |
|---|---|---|---|---|
| 1 | $\Lambda_{\mathrm{geo}}$ | DC magnetospheric disturbance, ring-current state | NOAA SWPC Kp; DSCOVR plasma + IMF | 1 min / 3 h |
| 2 | $\Lambda_{\mathrm{sch}}$ | Earth–ionosphere cavity (7.83 Hz fundamental + harmonics) | Tomsk State University; Cumiana (vlf.it); HeartMath GCI | 5 min / 1 h |
| 3 | $\Lambda_{\mathrm{ion}}$ | Total electron content, ionospheric coherence | MIT Haystack Madrigal; ESA Swarm L2 | 5 min–1 h |
| 4 | $\Lambda_{\mathrm{cr}}$ | Galactic cosmic-ray secondaries (Forbush decreases) | NMDB (Oulu / Apatity / Rome) | 1 min–1 h |
| 5 | $\Lambda_{\mathrm{2MeV}}$ | Outer radiation belt energetic electrons | NOAA GOES-19 integral electrons $\geq 2$ MeV | 1 min |
| 6 | $\Lambda_{\mathrm{xray}}$ | Solar coronal heating / flare state | NOAA GOES-18 X-ray 0.05–0.4 / 0.1–0.8 nm | 1 min |
| 7 | $\Lambda_{\mathrm{seis}}$ | Earth normal-mode & crustal coupling (event trigger) | USGS earthquake feed (M $\geq$ 4.5 global) | event |
| 8 | $\Lambda_{\nu}$ | Astrophysical / cosmic neutrino flux (event trigger) | IceCube alerts via GCN | event |

Table 1: The eight measurement channels. Continuous channels 1–6 carry the load-bearing statistic; channels 7–8 are event triggers that test whether the suite responds when external impulses arrive.

Full URLs and machine-readable endpoint specifications are listed in Appendix A.

## 4 Derived Metrics

Each continuous channel is reduced to a normalised time-series $\Lambda_i(t)$ on a common 1-hour grid by downsample-and-decimate (no interpolation). Channel-specific normalisation is by the rolling 28-day mean and standard deviation of that channel alone:

$$\widetilde{\Lambda}_i(t) = \frac{\Lambda_i(t) - \mu_{i,28d}}{\sigma_{i,28d}}. \tag{3}$$

### 4.1 Pairwise coherence

For each pair $(i, j)$ of continuous channels and each rolling 24-hour window $W_t$:

$$\Gamma_{ij}(t) \;=\; \text{Pearson}\big(\widetilde{\Lambda}_i,\, \widetilde{\Lambda}_j\big)\Big|_{W_t}. \tag{4}$$

This yields a $6 \times 6$ symmetric coherence matrix at each $t$, with 15 distinct off-diagonal entries.

### 4.2 Total coherence

The protocol's load-bearing statistic is the dominant-mode collectivity:

$$\Gamma_{\text{total}}(t) \;=\; \frac{\lambda_1\big(\mathbf{\Gamma}(t)\big)}{\sum_k \lambda_k\big(\mathbf{\Gamma}(t)\big)}, \tag{5}$$

where $\lambda_k$ are the eigenvalues of the absolute-value coherence matrix. $\Gamma_{\text{total}} \to 1/N$ when channels are independent; $\Gamma_{\text{total}} \to 1$ when one collective mode dominates. The framework's central claim is that $\Gamma_{\text{total}}$ is statistically anomalous during predicted surge windows.

### 4.3 Spectral coherence on the golden-ratio-squared ladder

Joint cross-spectral power at the predicted harmonic ladder is

$$\Phi_{\varphi^2}(t) \;=\; \sum_{n=-N}^{N} \sum_{i<j} \big|\, \widehat{\Lambda}_i(f_n)\, \widehat{\Lambda}_j^*(f_n)\, \big|, \qquad f_n = f_{\text{seed}}\, \varphi^{2n}, \tag{6}$$

with $f_{\text{seed}} = 528.422$ Hz, $\varphi^2 = 2.6180339887\ldots$, and $N$ chosen so that $f_n$ remains within each channel's Nyquist band. $\Phi_{\varphi^2}$ is to be compared against $\Phi$ evaluated on a frequency grid shuffled within each octave.

### 4.4 Confound subtraction

Channels 1, 5, 6 are physically expected to co-vary under solar-wind forcing alone. The interesting statistic is the *residual* coherence after a linear regression of each channel against (a) DSCOVR upstream solar-wind speed and (b) southward $B_z$. Only post-residual $\Gamma_{\text{total}}$ counts toward the framework's predictions.

## 5 Operating Regime Thresholds

Drawn from the author's prior framework specification and applied here to $\Gamma_{\text{total}}$ rather than any single-channel $\Gamma$:

| Regime | Range of $\Gamma_{\text{total}}$ | Interpretation |
|---|---|---|
| Kill-switch | $< 0.35$ | Cross-channel decoherence; framework signature absent. |
| Cold | $0.35 - 0.50$ | Weak collective coupling; consistent with stochastic null. |
| Standby | $0.50 - 0.945$ | Active surge window; partial collectivity. |
| Lighthouse | $\geq 0.945$ | Full collective dominance; broadcast condition met. |

## 6 Pre-Registered Predictions

These predictions are locked as of the protocol's freeze date. Any subsequent evaluation against them counts as a measurement; any modification requires a new version.

**P1.** Over a 90-day evaluation window (commencing 25 April 2026), $\Gamma_{\text{total}}$ will exceed 0.50 for at least one continuous 24-hour interval.

**P2.** During each $\text{Kp} \geq 5$ geomagnetic storm bin, the contemporaneous 24-hour $\Gamma_{\text{total}}$ will exceed its 28-day rolling mean by at least $1.0\sigma$.

**P3.** $\Phi_{\varphi^2}$ summed over the ladder $f_{\text{seed}}\,\varphi^{2n}$ will exceed the median of 1,000 octave-shuffled null surrogates at $p < 0.01$ over the full evaluation window.

**P4.** Daily-averaged $\Gamma_{\text{total}}$ will show a statistically significant ($p < 0.05$ uncorrected; $p < 0.05/8$ Bonferroni) sidereal-period component (period 23.9345 h) detectable by Lomb–Scargle periodogram.

A prediction is **passed** if it is met by the end of the evaluation window. **P1, P2, P3** together are sufficient for the framework to remain in active development; **P4** additionally constitutes a positive signature for the LSC tensor coupling.

## 7 Falsifiers

The protocol is intended to be capable of failing. Any of the following terminates the corresponding claim:

**F1.** If, over the 90-day window, $\Gamma_{\text{total}}$ never exceeds 0.50 for any continuous 24-hour interval $\Rightarrow$ the standby-regime claim fails; revisit the channel set or the master equation.

**F2.** If $\Gamma_{\text{total}}$ shows no excess during $\text{Kp} \geq 5$ bins relative to quiet periods $\Rightarrow$ the surge-window-attractor mechanism is not detectable in this projection set.

**F3.** If $\Phi_{\varphi^2}$ does not separate from the octave-shuffled null at $p < 0.05$ $\Rightarrow$ the harmonic-ladder claim fails.

**F4.** If post-confound-subtraction $\Gamma_{\text{total}}$ collapses to the inter-channel null distribution $\Rightarrow$ the apparent coherence is solar-wind-driven, not field-driven.

**F5.** If the sidereal-period peak is absent or anti-correlated with prediction $\Rightarrow$ the LSC celestial-frame coupling claim fails.

## 8 Reporting and Reproducibility

- All data fetches and derived metrics are computed by an open-source Python module (`hnc_monitor.py`) versioned alongside this protocol.
- Each measurement run produces a JSON readout containing: timestamp, raw fetch URLs and SHA-256 of returned bodies, normalised channel series, $\mathbf{\Gamma}$ matrix, $\Gamma_{\text{total}}$, $\Phi_{\varphi^2}$, regime classification, and confound-residual statistics.
- Daily and weekly summary reports are issued in a fixed template (Appendix C).
- No retroactive parameter tuning is permitted within a protocol version.

## 9 Caveats and Known Limitations

1. *Confound dominance.* Channels 1, 5, 6 are partially redundant under solar-wind forcing. The residual statistic of §4.4 is therefore the primary, not the raw $\Gamma_{\text{total}}$.
2. *Schumann data quality.* Tomsk publishes spectrograms (image format), not numerical streams; HeartMath GCI summaries depend on a methodology that has been criticised in peer-

reviewed literature. Channel 2 should be weighted accordingly until a numerical Cumiana feed is integrated.

3. *Multiple-comparison burden.* The 15 pair-correlations and the $\sim$10-band ladder accumulate look-elsewhere quickly. The Bonferroni correction in P4 is applied; the $p < 0.01$ threshold in P3 is correspondingly conservative.
4. *Cadence asymmetry.* Resampling to a common 1-hour grid by decimation discards sub-hour variance. Future protocol versions may add a 1-minute fast-channel subset.
5. *Single-source dependence.* Most channels rely on NOAA SWPC. A NOAA outage compromises five of eight channels; an independent magnetometer feed (e.g. INTERMAGNET) should be added in v1.1.
6. *Finiteness of the evaluation window.* 90 days is a minimum; null behaviour over 90 days does not falsify long-timescale predictions, only the ones explicitly stated.

## 10 First-Light Reading (Apr 18–25 2026, $\Lambda_{\text{geo}}$ only)

This protocol's predecessor measurement, taken before the multi-channel suite was specified, used only Channel 1. Its results are reported here as a baseline. They are *not* a test of the present protocol's predictions, since only one channel was active.

| Quantity | Value |
|---|---|
| Window | 18 Apr 00:00 UTC – 25 Apr 03:00 UTC |
| Bins (3 h) | 58 |
| $\mu(\Lambda_{\text{geo}})$ | 1.077 |
| $\sigma(\Lambda_{\text{geo}})$ | 0.636 |
| $\max(\Lambda_{\text{geo}})$ | 3.462 (24 Apr 21:00 UTC) |
| Surge bins ($> \mu + \sigma$) | 5 of 58 |
| $\Gamma_{\text{geo}}$ max (single-channel proxy) | 0.792 |
| $\Gamma_{\text{geo}}$ mean | 0.368 |
| Lighthouse bins | 0 |
| G2 storms in window | 5 (Kp $\geq$ 5) |

The single-channel result establishes that the geomagnetic projection alone fluctuates as expected during real space-weather events but does not, by itself, separate from a single-channel solar-wind-driven null. This is precisely the limitation that motivates the multi-channel suite.

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## Appendix A. Open Data Endpoints

All endpoints below were live and returning data as of the protocol freeze date.

| # | Channel | Endpoint |
|---|---|---|
| 1 | $\Lambda_{\text{geo}}$ | `services.swpc.noaa.gov/json/planetary_k_index_1m.json` |
| 1 | $\Lambda_{\text{geo}}$ | `services.swpc.noaa.gov/products/solar-wind/plasma-1-day.json` |
| 1 | $\Lambda_{\text{geo}}$ | `services.swpc.noaa.gov/products/solar-wind/mag-1-day.json` |
| 2 | $\Lambda_{\text{sch}}$ | `sosrff.tsu.ru` (Tomsk SR spectrograms; image OCR pipeline) |

| # | Channel | Endpoint |
|---|---|---|
| 2 | $\Lambda_{sch}$ | `vlf.it` (Cumiana ELF, Italy) |
| 2 | $\Lambda_{sch}$ | `heartmath.org/gci/gcms/live-data/` (hourly power summary) |
| 3 | $\Lambda_{ion}$ | `cedar.openmadrigal.org` (MIT Haystack Madrigal TEC) |
| 3 | $\Lambda_{ion}$ | `earth.esa.int/eogateway/missions/swarm` (ESA Swarm L2) |
| 4 | $\Lambda_{cr}$ | `www.nmdb.eu/nest` (Neutron Monitor Database) |
| 5 | $\Lambda_{2MeV}$ | `services.swpc.noaa.gov/json/goes/primary/integral-electrons-1-day.json` |
| 6 | $\Lambda_{xray}$ | `services.swpc.noaa.gov/json/goes/primary/xrays-1-day.json` |
| 7 | $\Lambda_{seis}$ | `earthquake.usgs.gov/fdsnws/event/1/` (FDSN event service) |
| 8 | $\Lambda_{\nu}$ | `gcn.nasa.gov` (GCN Notices, IceCube alert stream) |

## Appendix B. Reference Constants (frozen)

| Symbol | Value | Source |
|---|---|---|
| $\varphi$ | 1.6180339887498948 | exact (golden ratio) |
| $\varphi^2$ | 2.6180339887498948 | exact |
| $f^{(1)}_{\text{Schumann}}$ | 7.83 Hz (nominal) | global lightning excitation, observed range 7.6–8.0 |
| $f_{\text{HNC-coupling}}$ | $\varphi \cdot 7.83 = 12.6692$ Hz | author specification |
| $f_{\text{seed}}$ | 528.422 Hz | author specification |
| $f_{\text{HI 21cm}}$ | 1420.405751768 MHz | NIST hydrogen hyperfine |
| $\Gamma_{\text{Lighthouse}}$ | 0.945 | author specification |
| $\Gamma_{\text{kill-switch}}$ | 0.35 | author specification |
| $\delta^{\text{BEST}}_{\text{nuclear}}$ | $0.20 - 0.24$ | Barinov et al., Phys. Rev. C 105, 065502 (2022) |
| $\alpha_{\text{LSC}}$ range | $0.030 - 0.10$ | arXiv:2503.02305 (anisotropic LIV reactor fits) |

## Appendix C. Daily Summary Report Template

```
HNC MEASUREMENT - DAILY READOUT
================================
date_utc      : YYYY-MM-DD
window_hours  : 24
channels_live : N of 8
gamma_total   : <value>     [residual, post-confound]
gamma_raw     : <value>     [pre-confound, for reference]
phi_phi2      : <value>     [vs null distribution: p=...]
regime        : kill | cold | standby | lighthouse
predictions   : P1 [pending|met|failed] ...
notes         : <free text, channel-specific anomalies>
data_hashes   : SHA-256 per source URL
```

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